Bhopal AQI Prediction using Machine Learning

1. Introduction

Air pollution is a major environmental and public health issue, especially in urban areas. The Air Quality Index (AQI) is used to measure the level of pollution and its impact on human health.
This project focuses on predicting the AQI for Bhopal using machine learning techniques based on historical pollution data.

2. Problem Statement

The goal of this project is to build a model that can predict AQI values based on pollutant concentrations such as PM2.5, PM10, NO_2, CO, etc.

Accurate AQI prediction helps in**:-**

- Monitoring pollution levels
- Taking preventive measures
- Raising public awareness

3. Dataset

The dataset used contains air quality data for multiple Indian cities, including pollutant levels and AQI values.

Source: Kaggle (Air Quality Data in India)

Data includes**:-**

- Pollutants (PM2.5, PM10, NO, NO2, NOx, NH3, CO, SO2, O3)
- AQI values
- Date and city

For this project, only **Bhopal data** was filtered and used.

4. Data Preprocessing

The following steps were performed:-

- Converted date column to datetime format
- Sorted data chronologically
- Removed unnecessary columns (Benzene, Toluene, Xylene)

Handled missing values:-

- Dropped rows with missing AQI
- Forward-filled remaining values
- Removed rows with insufficient data after feature creation

5. Feature Engineering

To improve model performance, additional features were created**:-**

Temporal Features:-

Day of week
Month
Weekend indicator

Lag Features:-

AQI lag 1 day
AQI lag 2 days

AQI lag 3 days
AQI lag 7 days

These features help capture trends and time-based patterns in pollution data.

6. Model Selection

Initially, Linear Regression was tested but performed poorly.
A **Random Forest Regressor** was chosen because:

- It handles non-linear relationships
- Works well with tabular data
- Provides better accuracy

7. Model Training

Dataset split: 80% training, 20% testing (chronological split)
Model trained using:

n_estimators = 100
random_state = 42

8. Model Evaluation

The model was evaluated using:

-Mean Absolute Error (MAE): 8.59
-R^2 Score: 0.74

These results indicate good predictive performance.

9. Implementation (UI)

A user-friendly interface was built using Streamlit.

Features of the UI:-

- Input pollutant values
- Option to use sample data
- Predict AQI instantly
- Display AQI category (Good, Moderate, etc.)
- Visual AQI meter (progress bar)

10. Challenges Faced

- Handling missing data in the dataset
- Ensuring correct feature alignment between training and UI
- Managing time-series nature of data
- Fixing errors during model saving and UI integration

11. Key Learnings

- Importance of data preprocessing
- Feature engineering improves model performance significantly
- Random Forest is effective for real-world datasets
- Building UI makes ML models more practical and usable

12. Future Improvements

- Use real-time air quality data
- Deploy the application online
- Improve accuracy using advanced models (e.g., XGBoost)

- Extend support to multiple cities

13. Conclusion

This project successfully demonstrates how machine learning can be used to predict AQI using historical data. By combining data analysis, feature engineering, and model deployment, a complete end-to-end solution was developed.